

# Nate Cirino

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Available May 2027

## EDUCATION

### Northeastern University

*B.S. in Computer Science, Systems Concentration*

May 2027 | MGPA: 3.6/4.0

*Boston, MA*

**Coursework:** Foundations of Software Engineering, Object-Oriented Design, Algorithms & Data Structures, Computer Systems, Computer Networks, Systems Security, Cybersecurity, Foundations of Data Science, Digital Design & Computer Architecture

## EXPERIENCE

### Wayfair

*Software Engineering Co-op*

Jul 2026 – Present

*Boston, MA*

- Designing and implementing a supplier-agnostic e-label integration API that unifies 4 vendor systems behind one interface, enabling new outlet launches without per-supplier rework
- Coded, tested, and shipped customer-facing POS checkout flows: in-store scan-to-basket customer switching and live associate-entry customer sync
- Support the front-of-house application stack across all Wayfair retail brands (AllModern, Birch Lane, Joss & Main, Perigold), gathering requirements from store associates to drive business outcomes

### Wolters Kluwer

*Software Engineer*

Jul 2025 – Jun 2026

*Boston, MA*

- Re-engineered Relay JMS broker serialization with json-io across 20+ integrated microservices, eliminating timeouts at production scale
- Delivered application security by implementing OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians
- Built and unit-tested a Java Spring microservice for CME credit issuance inside a clinician-facing workflow
- Integrated Expert AI into CME accrual/redemption microservices, expanding decision support to 250K+ clinicians
- Tuned search and filtering configuration across 20+ microservices, improving clinical query relevance at scale

### SculptAI

*Software Architect*

Jan 2025 – Aug 2026

*New York, NY*

- Supported and monitored production cloud infrastructure at 99.99% uptime across 1–2M monthly model calls for 2K paying users via AWS WAF, Datadog, and Kubernetes
- Architected a Flask backend and retrieval pipeline on AWS Bedrock delivering personalized workout recommendations
- Migrated the retrieval stack from LangChain to Vespa to zero-ops Bedrock, improving quality while reducing operational overhead
- Restructured the engineering org at a \$1M-funded startup into DevOps, development, and agentic workflow teams running Agile delivery
- Continue as technical advisor on architecture and integration direction

### Chevron New Energies

*Software Engineer Intern*

May 2024 – Aug 2024

*Houston, TX*

- Built RESTful APIs and data integration pipelines using Azure Data Factory, improving product efficiency by 40%
- Configured CI/CD workflows in Azure DevOps and Kubernetes, enabling automated deployment, testing, and rollback

## PROJECTS

### Sentinel | *Next.js, FastAPI, Claude, pgvector, OpenTelemetry, Docker*

Apr. 2026 – Present

- Built an AI support agent that autonomously triages application incidents using Prometheus, Tempo, and Loki telemetry
- Built a pgvector retrieval layer over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy under a formal QA suite
- Gated all remediation behind human approval; running live in SculptAI production triaging real incidents

### Kepler | *Rust, C++, Raft, LSM-tree, MVCC, Tokio*

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree engine
- Implemented Raft election, log replication, and membership changes; storage: memtable, WAL, leveled SSTables

- Added MVCC transaction layer providing snapshot isolation and linearizable reads across a multi-node cluster
- Built single-tier compaction with full SSTable rewrite every 4 flushes; inline, writer-blocking execution

**NOAA AUV Sonar Detection** | *Python, PyTorch, AWS Neuron SDK, SageMaker* April 2025 – July 2025

- Trained CNN models on thermal sensor spectrograms to detect novel aquatic signatures for NOAA AUVs
- Refined architecture and loss formulation to address class imbalance and sensor noise, improving model detection
- Designed an AWS deployment path for onboard inference under hard real-time constraints

## SKILLS & HONORS

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**Languages/Tools:** Java, Python, TypeScript, JavaScript, C, C++, Rust, Bash, SQL/PostgreSQL, Azure, AWS, Docker, Kubernetes, Terraform, Jenkins, Buildkite, Groovy, GraphQL, Linux

**Frameworks & Libraries:** Java Spring, Flask, FastAPI, Node.js, Express, React, Next.js, PyTorch, TensorFlow, Pandas

**AI / Agentic Tooling:** Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

**Honors:** Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor